

Dietary Approaches to Kidney Diseases

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KEY POINTS

- Diet is an underused strategy by clinicians in their goal to prevent or slow progression of chronic kidney disease (CKD) to end-stage kidney disease (ESKD) and in the management of CKD-related complications.
- Diet is a cornerstone strategy in the prevention and management of the two syndromes that are the major contributors to CKD in developed societies, diabetes mellitus and hypertension.
- Predominantly plant-based diets are associated with reduced overall mortality risk in CKD, an important consideration for clinicians given that CKD patients suffer higher mortality than patients without CKD.
- Diets low in acid production and that promote a gut microbiome with low levels of nephrotoxic metabolites are associated with reduced risk for initiation and progression of CKD.
- Diet modifications can reduce uremic complications due to accumulation of nitrogenous wastes. Such modifications include limiting the amount of, and focusing on the character of, ingested dietary protein.
- Predominantly plant-based diets appear to have an important role in CKD management, and CKD patients with reduced but comparatively well-preserved glomerular filtration rate appear able to tolerate the increased potassium load of predominantly plant-based diets better than once thought.
- Public policy approaches to implementation of “healthy” diet components for populations might prevent CKD and reduce its incidence, particularly in high-risk populations, in addition to provision of these diet components to individuals by their clinicians.

INTRODUCTORY POINTS

DIET IS A KEY COMPONENT OF THE MANAGEMENT OF CHRONIC KIDNEY DISEASE

Analysis of US data showing that diet is the single largest CKD-related risk factor for death and disability in individuals

with chronic kidney disease (CKD) show the importance of diet in CKD management.¹ Furthermore, a “healthy” diet in CKD patients is associated with reduced mortality.^{2,3} These data support that diet can enhance the length and quality of life of individuals with CKD and should encourage clinicians to know and recommend food patterns which accomplish these goals. Additional reasons for clinicians to

consider diet as the cornerstone of CKD management include the following:

- Diet is a cornerstone component of the management of diabetes, the major cause of CKD in developed societies.⁴ Individuals with diabetes-related CKD must have their diabetes properly managed along with their CKD. Importantly, the “healthier” (we will subsequently detail the components of a “healthy” diet) the diet in CKD due to type 2 diabetes (T2D) the lower the mortality and the lower the risk for CKD progression.⁵
- Diet is recommended as first-line management of hypertension,⁶ a comorbid condition present in more than 90% of individuals with CKD.⁴ Consequently, diet will help control hypertension in CKD and long-term studies support that better hypertension control yields lower mortality in CKD.^{7,8}
- Diet helps control some of the metabolic complications of CKD.
- Diet also helps ameliorate some of the organ dysfunction associated with CKD.
- Properly applied, diet can limit body accumulation of waste products ordinarily excreted by the kidney that cause harm, thereby delaying the need for kidney replacement therapy in the form of dialysis or kidney transplantation. The accompanying imbalance in acid-base, fluid, electrolytes, and mineral bone disease decreases the quality of life of patients with CKD. Ameliorating these metabolic complications by skillful application of effective food patterns that successfully delay the need for kidney replacement therapy can improve patient life quality and reduce societal costs for CKD management.
- Diet might also be used to help slow CKD progression, as will be discussed.

Together, these data support that diet in CKD plays an important role in its prevention, its progression once it is established, and in management of its complications. Although the focus of our discussion will be strategies and tactics available to clinicians caring for CKD patients, the data will support that changes in food policy will facilitate some interventions, particularly when considering strategies for CKD prevention. The data to be discussed support diet as the foundational, “food first”⁹ approach to the management of individuals with CKD as well as those at risk for it, with pharmacologic therapy as adjunctive to diet. This is the reverse of the current approach in which pharmacologic therapy is foundational to CKD management. We will begin our discussion with the contributions of diet to the major disease entities responsible for CKD, diabetes and hypertension, which account for 72% of incident end-stage kidney disease (ESKD) cases in the US.⁴

DIETARY CONTRIBUTIONS TO THE MAIN SYNDROMES WHICH CAUSE CKD: DIABETES AND HYPERTENSION

DIABETES

Diabetes mellitus remains the single largest contributor to ESKD in the US with T2D being responsible for greater than

90% of the cases and type 1 (T1D) contributing the remainder.^{4,10} Although US incidence of T2D is decreasing, its prevalence is increasing,¹¹ likely due to T2D individuals living longer which unfortunately exposes them to greater risk of developing CKD. Ten years following T2D diagnosis, 5.3% had macroalbuminuria (i.e., greater than 300 mg of daily urine albumin excretion [4]), indicative of established nephropathy.¹² Unfortunately, many T2D patients already have signs of CKD at the time of diagnosis,¹³ possibly because they had experienced an indeterminate period of the abnormal metabolic state known as “pre-diabetes” which can be associated with signs of kidney injury, like albuminuria.¹⁴ More concerning, about one third of US adults have “pre-diabetes”¹¹ and such individuals have a higher prevalence of reduced eGFR compared to the general population.¹⁴ In the past quarter century, both cardiovascular and kidney disease have become highly prevalent in adults with prediabetes, irrespective of the definition used to define prediabetes.¹⁴ It is therefore important to consider dietary approaches to CKD, particularly strategies intended to prevent CKD onset and its progression, within the context of dietary contributions to this major health condition that leads to CKD in developed societies.

Dietary contributions to T2DM. Overweight is the single most important modifiable risk factor for T2DM¹⁵ and so the dietary factors that contribute to it provide the context for diabetes-related CKD. Observational studies support that adherence to a “healthy” as opposed to an “unhealthy” diet, including the “Mediterranean” diet that emphasizes fresh fruits and vegetables, olive oil, fish as the source of animal protein, and de-emphasizes simple sugars and red meat as a protein source, is associated with reduced T2D risk.^{16,17} Dietary components that appear to influence T2D risk include:

- *Character of ingested carbohydrates.* High intake of the indigestible component of complex carbohydrate known as fiber is associated with reduced T2D risk.^{18,19} Increased fiber intake enhances the growth of gut bacteria whose fermentation of the ingested fiber produces short chain fatty acids that promote improved glycemic control.²⁰ Whole grain foods are particularly high in fiber. Fruits and vegetables, in general, are fiber sources. On the other hand, chronically high ingestion of foods with greater propensity to increase blood glucose levels when ingested (i.e., foods with a high glycemic index such as simple sugars) is associated with increased T2D risk.^{21–23} The glycemic index is a relative ranking of carbohydrate in foods according to how they affect blood glucose levels. It is a number from 0 to 100, with pure glucose arbitrarily given the value of 100. Carbohydrates with a low glycemic index value (55 or less) are more slowly digested, absorbed and metabolized and cause slower rise in blood glucose, often to lower levels.
- *Character of ingested fat.* Although high intake of saturated fat appears not to affect the risk for T2D,²⁴ high intake of the polyunsaturated fats omega 6^{25,26} and omega 3²⁷ appear to reduce the risk for T2D.
- *Character of ingested protein.* High intake of animal-based protein is associated with increased risk for T2D,^{28,29} particularly increased intake of red meat.^{28–31} On the other hand, diets proportionally high in plant-based protein are associated with reduced T2D risk.³²

Table 60.1 Dietary Approaches to Reduce Incidence of Type 2 Diabetes

Strategy	Tactic
Emphasize carbohydrates that promote healthy glucose metabolism	- Promote foods with dietary fiber - Whole grains - Fruits and vegetables - Minimize foods with high glycemic index - Simple sugars - Fructose
Emphasize fats that promote healthy fat metabolism	Promote polyunsaturated fats, including omega 6s and omega 3s
Emphasize dietary protein that is less likely to increase insulin resistance	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Reduce obesity risk	- Promote a healthy caloric intake - Maximize foods with high nutrition/low calorie ratio like fruits and vegetables - Minimize processed foods - Promote omega 3 polyunsaturated fats

- *Dietary effects on endogenous acid production.* Animal-based protein is metabolized to yield acid but most plant-based protein is metabolized to yield base or is metabolized to yield neither.³³ In addition, a high salt (NaCl) diet increases endogenous acid production.³⁴ Whether ingested diets are acid-producing or base-producing is determined to large degree by the balance of acid-producing and base-producing dietary components. High acidogenic diets increase insulin resistance,³⁵ their chronic ingestion is associated with increased T2D risk,^{36–38} and insulin resistance in T2D is associated with hypertension and microalbuminuria.³⁶

Together, these data guide dietary strategies to prevent diabetes, particularly in individuals and populations at risk (Table 60.1). In addition, they help guide dietary management of CKD patients with concomitant diabetes.

HYPERTENSION

Hypertension is the second largest contributor to ESKD in the US and is a comorbid factor in more than 90% of patients with CKD.⁴ As discussed for diabetes, it is important to consider dietary approaches to CKD, particularly strategies intended to prevent CKD, its progression, and the effect of CKD on mortality, within the context of dietary contributions to hypertension. This is particularly important given that better blood pressure control in CKD is associated with lower mortality.^{7,8}

Dietary contributions to hypertension. As described for diabetes, adherence to a "healthy" as opposed to an "unhealthy" diet is associated with lower incidence of

Table 60.2 Dietary Approaches to Reduce Incidence of Hypertension

Strategy	Tactic
Emphasize a "healthy" profile of electrolyte intake	- Minimize NaCl ("salt") - Fresh rather than processed foods - Fresh fruits and vegetables - Promote potassium intake as tolerated - Fruits and vegetables - Promote magnesium intake as tolerated - Fruits and vegetables
Emphasize proteins that promote healthy blood pressure	- Promote plant-based foods - Minimize animal-based foods
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Reduce obesity risk	- Promote a healthy caloric intake - Maximize foods with high nutrition/low calorie ratio like fruits and vegetables - Minimize processed foods

hypertension in populations.^{39,40} Other observational studies support that population adherence to the previously described Mediterranean diet is associated with lower blood pressure.⁴¹ Furthermore, current guidelines recommend the Dietary Approaches to Stop Hypertension (DASH) diet as first-line therapy for hypertension⁶ because it effectively reduces blood pressure.⁴² This diet emphasizes fruits and vegetables, whole grains and nuts, and limits consumption of animal-based protein, such as red meat, in preference for moderate intake of fish and poultry. The DASH diet reduces blood pressure in hypertensive patients.⁴² Dietary components that appear to contribute to hypertension risk are summarized in Table 60.2, and include:

Electrolyte intake. Multiple studies highlight the important role of sodium (NaCl or "salt") in the initiation and maintenance of hypertension, noting that hypertension is rare in societies with very low dietary sodium intake, even in individuals of advanced age.⁴³ In addition, there is a direct relationship between population sodium intake and blood pressure,^{43,44} and dietary sodium restriction reduces blood pressure in patients with hypertension.⁴⁵ As noted, the DASH diet reduces blood pressure⁴² but sodium restriction further reduces blood pressure in patients eating this diet.⁴⁶ By contrast, high dietary sodium appears to abrogate the blood pressure-lowering effects of a "healthy diet".⁴⁷ On the other hand, urine potassium excretion, which is reflective of dietary potassium intake, is inversely associated with blood pressure.^{48,49} Dietary potassium appears to have a number of blood pressure-lowering benefits, including facilitating urine sodium excretion.⁵⁰ The dietary sodium and potassium combination that appears to reduce hypertension risk best is low sodium combined with high potassium intake.^{48,49} In addition, some studies suggest that dietary magnesium intake is inversely associated with hypertension risk^{51,52} and some others show that magnesium

supplements can slightly decrease blood pressure in those with hypertension.⁵³

- A plant-based diet, including fresh fruits and vegetables is naturally low in sodium and high in potassium,⁴⁸ and is a good source of magnesium.⁵²
- *Character of ingested protein.* Components of “unhealthy” diets, which have been associated with augmented hypertension risk, include increased intake of animal-based protein such as cheese and processed red meat.⁵⁴ In contrast, food components associated with lower hypertension risk include plant-based protein, like fruits and vegetables.^{55,56} Epidemiologic studies show that strict vegetarians have a lower incidence of hypertension than those who are not⁵⁷ and that the blood pressure of vegetarians is less likely to increase with aging as is typically the case in most individuals eating the standard diets of developed societies.⁵⁸ Such studies suggest that reduced dietary intake of particular components of our standard diet was not the responsible factor for this blood pressure difference but that lower blood pressure in vegetarians was due instead larger intake of potentially beneficial nutrients from the increased intake of vegetables (and fruits).⁵⁹ In support of this contention, supplemental addition of fruits and vegetables to the Mediterranean diet, one that is similar in make-up to the DASH diet and is already higher in vegetables than diets typical of developed societies, was associated with a lower risk for development of hypertension.⁶⁰ Furthermore, reduction of red meat consumption and substitution with vegetable protein reduced blood pressure in pre-hypertensive, post-menopausal women.⁶¹
- *Dietary effects of endogenous acid production on hypertension.* As indicated, the body metabolizes animal-based protein in a way that increases endogenous acid production whereas most plant-based protein is metabolized in a way that either decreases endogenous acid production or has no net effect on it.³³ In addition, as indicated, high salt (NaCl) diets increase endogenous acid production.³⁴ Indeed, dietary NaCl intake has approximately 50% to 100% of the acidosis-producing effect of the diet net acid load.³⁴ Studies in children⁶² and adults^{63,64} support that diets that increase endogenous acid production increase blood pressure. Because dietary sodium increases,³⁴ dietary potassium decreases,³³ animal-based protein increases,³³ and plant-based protein decreases³³ endogenous acid production, dietary effect on endogenous acid production might be a common mechanism through which these and possibly other dietary components influence blood pressure and the risk for hypertension.
- *Obesity.* There are, of course, strong dietary contributions to obesity and population studies show that obesity is associated with increased risk for hypertension.^{65,66} In addition, weight loss in obese patients with hypertension reduces their blood pressure.⁶⁷

Strategies to reduce CKD incidence in populations can begin by reducing the incidence of its two major causes, diabetes (particularly T2D) and hypertension. Data presented support strong dietary contributions to both and suggest that dietary strategies instituted for individual at-risk patients and, more broadly, populations as a whole, hold promise to reduce CKD incidence through reducing the incidence of

diabetes and hypertension. Such interventions might be more effectively instituted at an overall population, rather than on an individual, level, particularly when focused on population segments at particularly high CKD risk. Furthermore, because diabetes is the single largest contributor to CKD in developed societies and because more than 90% of CKD patients have hypertension, these data support effective dietary strategies in individual CKD patients to manage concomitant diabetes and/or hypertension in CKD patients.

DIETARY CONTRIBUTIONS TO CKD INCIDENCE AND PROGRESSION

DIETARY FACTORS ASSOCIATED WITH INITIATION OF CKD

Similar to what has been described for the two syndromes that are the main contributors to CKD, observational studies support that adherence to a “healthy” as opposed to an “unhealthy” diet is associated with reduced risk for CKD.^{68–71} Although the components of a “healthy” diet that mediate reduced CKD risk remain to be clearly identified, observations that adherence to the “Mediterranean” diet which emphasizes fresh fruits and vegetables, olive oil, and fish as the source of animal protein and de-emphasizes red meat and simple carbohydrates is associated with reduced CKD risk^{72–75} offers some dietary elements to consider.

Plant-based diets are high in fiber and such diets, particularly those with a high ratio of fiber to animal-based protein, are associated with reduced CKD risk,^{75,76} possibly in part because diets with a high proportion of animal-based protein are associated with high levels of gut-derived substances that are putative kidney toxins.⁷⁷ As indicated, most plant-based protein is metabolized to yield base after ingestion as opposed to animal-based protein which yields acid.³³ Low acid diets are associated with reduced, and high acid diets with increased, CKD risk.^{78,79} High intake of red meat, in particular, is associated with increased CKD risk.^{80,81} High dietary acid diets, caused largely by disproportionately high intake of animal-based protein, might contribute to kidney injury that initiates CKD by causing chronic hyperfiltration,⁸² a phenomenon linked to hemodynamically-induced kidney injury.⁸³

Other potential kidney protective benefits of the Mediterranean diet might derive from its oily fish and other plant-based foods which contain omega 3 polyunsaturated fatty acids, high intake of which has been associated with reduced CKD risk.⁸⁴ Omega 3 polyunsaturated fatty acids are also associated with reduced risk of obesity⁸⁵ and obesity itself is associated with increased CKD risk,^{86,87} even independent of diabetes and hypertension.^{88,89}

DIETARY FACTORS ASSOCIATED WITH PROGRESSION OF ESTABLISHED CKD

Individuals with albuminuria, even with initially normal eGFR, have increased risk for subsequent eGFR decline⁴ whether due to diabetes⁹⁰ or to nondiabetic CKD associated with primary hypertension.⁹¹ As has been described for diabetes, hypertension, and initiation of CKD, adherence to a “healthy” as opposed to an “unhealthy” diet is associated slower eGFR decline of established CKD.^{5,92,93} These “healthy” diets have

high content of plant-based protein^{68–71} and increasing the proportion of dietary plant-based protein decreases urine indices of kidney injury⁹⁴ and slows CKD progression.^{95–97} Because, as indicated, plant-based protein is largely metabolized to yield base, at least part of the benefit of plant-based diets might be their effect to reduce dietary acid^{33,94} because correction of metabolic acidosis^{96–99} and dietary acid reduction even without underlying metabolic acidosis¹⁰⁰ slows the rate of eGFR decline in CKD. Other mechanisms contributed by plant-based protein might be promotion of a gut microbiome that yields reduced metabolism of diet components delivered to the gut into kidney toxic substances.¹⁰¹ Diets with high intake of plant-based components are comparatively lower in sodium than diets high in animal-based components, particularly processed meats, and high sodium diets have been associated with progression of CKD.¹⁰²

POTENTIAL MECHANISMS FOR DIET-RELATED KIDNEY INJURY

- **Hyperfiltration.** When a substantial number of functioning nephrons is lost, remaining functioning nephrons increase filtration above that when they were part of a normal contingent of nephrons, a process known as hyperfiltration.⁸³ Animal models of CKD support that the high glomerular filtration pressure associated with hyperfiltration is injurious to kidneys in the long term and contributes to progressive GFR decline.¹⁰³ High intake of animal-based protein contributed to hyperfiltration with increased glomerular capillary pressure in these animal models of CKD and these diets high in animal-based proteins induced greater kidney injury in animals with reduced compared to those with intact nephron mass.^{103,104} Chronically high intake of animal-based dietary proteins also appears to cause hyperfiltration in humans.⁸² Hormonal systems that appear to contribute to hyperfiltration in these animal models of CKD include glucagon, insulin, insulin-like growth factor-1, angiotensin II, prostaglandins, and kinins.^{105–108}
- **Oxidative stress.** Oxidative metabolism yields molecules that are potentially injurious to surrounding tissue and innate systems normally remove or detoxify them. These oxygen-related molecules can cause tissue injury if their production exceeds normal detoxifying mechanisms or if the normal detoxifying mechanisms are compromised by a phenomenon known as oxidative stress. The reduced nephron mass animal model of CKD is associated with oxidative stress that is exacerbated by increased intake of animal-based dietary protein.^{109,110}
- **Inflammation.** Progressive nephropathy is characterized by inflammation, including fibrosis, particularly of the kidney tubule interstitium.¹¹¹ Diets high in animal-based protein increase kidney production of transforming growth factor β -1 and fibronectin, two substances that mediate kidney fibrosis.¹¹² Furthermore, high-intake of animal-based protein increased kidney levels of endothelin-1, aldosterone, and angiotensin II, each of which mediates progressive GFR decline and tubule-interstitial fibrosis in the reduced kidney mass model of CKD in animals. Conversely, increased intake of plant-based dietary protein ameliorated GFR decline and caused less tubule-interstitial fibrosis.^{113–116}
- **Acid-induced kidney injury.** As noted, high dietary acid is associated with increased CKD incidence^{78,79} but is also

associated with its progression.¹¹⁷ Animal studies show that high acid diets increase kidney levels of substances such as endothelin-1, aldosterone, and angiotensin II, all of which help mediate increased kidney tubule acidification in response to these acid diets^{118–120} but also cause kidney injury that contributes to progressive GFR decline.^{113–116,121} Correction of metabolic acidosis in CKD patients with amelioration of nephropathy progression is associated with reduced urine excretion of these substances.^{96,99,100} Together, these data support that high dietary acid causes acid-induced kidney injury through increased kidney levels of these, and possibly other, cytokines.

The data support that a plant-based diet is associated with less kidney injury or can ameliorate kidney injury by each of the diet-related mechanisms of kidney injury described.

DIETARY CONTRIBUTIONS TO INCREASED CKD-RELATED MORTALITY

Reduced eGFR is associated with increased mortality,¹²² due particularly to increased cardiovascular mortality,¹²³ and was associated with 4% of worldwide deaths in 2013.¹²⁴ The appearance of reduced GFR significantly shortens life expectancy in patients with diabetes compared those with diabetes and normal GFR.¹²⁵ Diet is the single largest CKD-related risk factor for death and disability¹ and a “healthy” diet is associated with reduced mortality in CKD.² Excess animal-based protein to fiber intake is associated with increased mortality in CKD patients¹²⁶ but increases in dietary fiber were associated with reduced CKD-related inflammation and mortality.¹²⁷

Together, these data support dietary strategies as front-line management of individuals at increased risk for or with established CKD in an effort to reduce their risk for developing CKD, for progression of established CKD, and to reduce the excess mortality associated with CKD.

COMPONENT CONSIDERATIONS IN THE DIETARY MANAGEMENT OF CKD

NUTRITIONAL COMPONENTS TO CONSIDER WHEN DESIGNING DIETARY MANAGEMENT OF CKD PATIENTS

- **Energy requirements.** Most studies support that energy requirements for CKD patients are not different from non-CKD patients but energy requirements did not decrease in response to decreased caloric intake as it did non-CKD patients,¹²⁸ suggesting that CKD patients do not respond normally to a reduction in caloric intake. Some studies support a slight increase in energy requirements in hemodialysis patients due to uremic factors that cause insulin resistance.¹²⁹ Although CKD patients might have steady-state levels of energy intake below the recommended 30–35 kcal/kg body weight/day,¹³⁰ most studies support the 30–35 kcal/kg body weight/day recommendation when prescribing a protein-restricted diet for kidney protection to avoid protein energy wasting.^{131,132} Carbohydrates can account for 5% to 15% of energy requirements for CKD patients¹³³ but practitioners must exercise caution when

including foods with high fructose content because it is associated with increased insulin resistance¹³⁴ and with increased all-cause and CVD mortality.¹³⁵ Additional reasons to limit or avoid dietary fructose as an energy source for CKD patients is that it increases serum uric acid levels,¹³⁵ which are associated with exacerbated nephropathy progression.^{136,137} Furthermore, lowering serum uric acid improves kidney outcomes and reduces the risk for cardiovascular events in CKD patients.¹³⁸

Protein requirements. Empiric observations support minimum dietary protein requirements as 0.6 g/kg of ideal body weight per day; adding 2 standard deviations for a general recommendation to theoretically encompass the dietary protein needs of 97.5% of individuals yields the standard recommendation of 0.75 g/kg bw/day¹³⁹ which is often rounded up to 0.8 g/kg/bw/day. This recommendation applies to both vegetarians and non-vegetarians.¹⁴⁰ These guidelines suggest that dialysis patients increase this amount to 1.2 g/kg bw/day because of their high protein catabolism and/or loss.¹⁴¹ Unlike that described for energy requirements, CKD patients can adapt to dietary protein restriction by reducing amino acid oxidation and protein degradation.¹⁴² Naturally occurring amino acids contain nitrogen, which when metabolized yield nitrogenous wastes that increase blood urea and possibly adversely affect kidney function.¹⁴³ Symptoms of uremia are associated with breakdown products of protein catabolism. Therefore, restriction in dietary protein can delay the need to start kidney replacement therapy despite a decline in GFR by ameliorating uremic symptoms. This confounds, by potentially introducing bias favoring dietary protein restriction, defining mechanisms that protect renal function directly. Dietary animal protein will reduce measured GFR, as defined by creatinine, unrelated to changes in true GFR.

Low protein diets (0.6–0.8 g/kg bw/day), sometimes supplemented with non-nitrogen ketoanalogues, have been associated with slower GFR decline^{143,144} and with lower levels of gut-derived uremic toxins¹⁴⁵ in non-dialysis CKD patients. Nevertheless, the largest study to date that examined the impact of total dietary protein restriction on nephropathy progression, the Modification of Diet in Renal Disease (MDRD) study, showed that GFR decline at three years was no different between subjects given a low protein diet (0.58 g/kg/day) and a usual protein diet (1.3 g/kg/day).¹⁴⁶ The MDRD study was the largest randomized clinical trial to test the hypothesis that protein restriction over a 2- to 3-year period slows the progression of chronic kidney disease. However, the primary results published in 1994 were not conclusive with regard to the efficacy of this intervention. Long-term follow-up of the cohort also remained inconclusive. Furthermore, despite the benefit of low protein diets to reduce the rate of GFR decline shown in some studies, some patients with intakes below 0.8 g/kg/bw/day, even the described minimum level of 0.6 g/kg/bw/day, develop protein energy wasting¹⁴⁷ and increased mortality.¹⁴⁴ Even so, some studies show that CKD patients can safely eat these diets without compromising nutritional status.^{143,148} Because of the described benefits of primarily plant-based diets, investigators have studied these diets

in CKD patients and show similar nutritional status in such patients compared to those ingesting similar amounts of a primarily animal-sourced protein diet.¹⁴⁹ Controversy still exists about the potential for a beneficial effect of reduction in dietary protein on the rate of progression of loss of kidney function. The level of residual kidney function at the time of initiation of therapy and the heterogeneity of etiologies of CKD in the populations tested are some of the factors that confound interpretation of collective studies.

Individuals living in developed societies typically ingest dietary protein in excess of requirements, averaging about 60% more,¹⁴⁰ and most of this ingested protein is animal-based.³³ As noted, animal-based protein increases intrinsic acid production when metabolized³³ and increased dietary acid has been associated with increased risk for CKD appearance^{78,79} and progression.^{96–100} Also as noted, animal-based dietary protein is associated with the production of high levels of gut-derived substances which are putative kidney toxins.⁷⁷ For these reasons, many guidelines recommend restricting dietary protein intake for those living in developed societies to that necessary to satisfy requirements, typically 0.7 to 0.8 g/kg of ideal body weight daily.¹³³

Ingested dietary protein that is not converted into protein is converted to urea, the major waste product derived from protein catabolism,¹⁵⁰ which can accumulate and contribute to uremic symptoms. Steady-state blood urea levels directly parallel levels of all nitrogen-containing waste products that contribute to uremic symptoms,¹⁵¹ so following urea levels is a helpful guide to the accumulation of these substances. Steady-state concentration of urea is a function of dietary protein intake that contributes to urea production of the comparatively small amount of urea degradation and extra-kidney clearance,¹⁵⁰ and of how well the kidney excretes urea, the latter being directly related to GFR and the tubular handling of urea. It follows that reducing dietary protein is an effective strategy to reduce urea and the associated blood levels of nitrogenous wastes in patients with declining GFR.

Metabolic acidosis, a comorbid condition that appears when eGFR is comparatively low (typically, less than 40% of normal),¹⁵² increases protein catabolism¹⁵³ mediated through the adenosine triphosphate-dependent ubiquitin-proteasome system.¹⁵⁴ Correction of the metabolic acidosis of CKD with alkali decreases protein catabolism, improves protein balance with increased muscle mass, and decreases induction of the ubiquitin-proteasome system.^{155,156} Metabolic acidosis also decreases protein anabolism, manifest by decreased albumin synthesis¹⁵⁷ that is improved with correction of the metabolic acidosis.¹⁵⁸ Because diets in developed societies are typically acidogenic³³ and can contribute to metabolic acidosis, particularly in individuals with low GFR,¹⁵⁹ modifying diet can help prevent or treat metabolic acidosis in CKD and thereby ameliorate its untoward effects on protein metabolism. Options to accomplish this modification include reducing the amount of animal-based protein in the diet and/or adding base-producing fruits and vegetables, the latter of which effectively treats metabolic acidosis in CKD.^{96,160}

Guidelines for dietary protein in CKD have most often focused on the amount of dietary protein but less so on character of the ingested protein. The 2013 Kidney Disease: Improving Global Outcomes (KDIGO)¹⁶¹ guideline 3.1.13 for CKD patients suggests “lowering protein intake to 0.8 g/kg body weight/day in adults with diabetes (“2C” level of strength, meaning that this is a suggestion and that the true effect of this suggestion might be substantially different from the estimated effect) or without diabetes (“2B” level of strength, meaning that this is a suggestion and that its true effect is likely to be close to the estimated effect but there is a possibility that it is substantially different) and GFR < 30 mL/min/1.73 m² (GFR categories G4-G5) with appropriate education.” The guidelines suggest that the recommended protein be of “high biologic value” but do not discuss if this includes plant-based protein. Guideline 3.1.14 reads “we suggest avoiding high protein intake (> 1.3 g/Kg/body weight/day) in adults with CKD at risk of progression” (“2C” level of strength, as explained above). The qualifications for these two suggestions reflect the need for additional research to develop more concrete dietary protein recommendations for patients with very low (< 30 mL/min/1.73 m²) GFR. In particular, further studies will elucidate importance of the protein’s character compared its amount, particularly with respect to the acid-producing or base-producing protein ability. In addition to the aforementioned potential advantages of plant-based compared to animal-based protein in CKD, carnitine and lecithin in animal-based protein are metabolized by gut microbes to trimethylamine, which in turn is metabolized by liver flavin monooxygenases to trimethylamine-N-oxide (TMAO).¹⁶² High blood levels of TMAO strongly correlate with CVD and are associated acute clinical events of CKD.^{162,163} Together, these data support considering the character, in addition to the amount, of dietary protein recommended for CKD patients, particularly plant-based protein for those able to tolerate the increased potassium that accompanies plant-based diets.

- **Sodium.** Adults in the United States consume an average of 3400-3600 mg (148-156 mmol) of sodium per day,^{164,165} an amount in great excess of dietary recommendations.¹⁴⁰ Processed foods comprise the majority of sodium consumed in the United States, which industry typically adds to increase shelf life of and enhance palatability for consumers.¹⁶⁶ Foods eaten outside of homes typically have even higher sodium content.¹⁶⁶ Higher dietary sodium in CKD is associated with higher blood pressure whereas its reduction is associated with reduced blood pressure and reduced albuminuria.^{167,168} In addition, high sodium diets are associated with CKD progression.¹⁰² Together, these data support some level of dietary sodium restriction below current dietary intake in CKD patients but there are not enough data to determine the optimal level of dietary sodium intake. The 2015 report of the American Dietary Guidelines Advisory Committee¹⁶⁹ recommends a daily sodium intake for adults of less than 2300 mg (100 mmol). The KDIGO recommendation (3.1.19)¹⁶¹ for adult CKD patients is a daily sodium intake of less than 2000 mg (90 mmol) unless otherwise contraindicated. Despite that this recommendation is of “1 C” quality (i.e., most clinicians

would agree but the true effect may be substantially different from the estimated effect), it seems to be reasonable until more definitive studies determine a more specific level of dietary sodium restriction.

- **Potassium.** Patients with CKD appear to benefit from diets high in plant-based components, including fruits and vegetables, as discussed but the high potassium content of these diets historically has made practitioners reluctant to prescribe them because of the reduced potassium excretory capacity of CKD patients with reduced GFR.¹⁶¹ Nevertheless, only limited data to support that dietary potassium restriction improves outcomes in CKD and most of the recommendations are opinion based. Relatedly, some reports question the historic admonition to limit intake of plant-based foods in CKD patients, including fruits and vegetables.¹⁷⁰ Enhanced kidney excretion mediated by increased serum potassium, increased serum aldosterone, and increased distal urine flow, and enhanced Na⁺-K⁺-ATPase activity can help maintain potassium balance in CKD.¹⁷¹ Patients with CKD stage 4 who had fruits and vegetables added to their ad lib diets increased urine aldosterone excretion and exhibited urine hormonal changes consistent with lower urine 11 β -hydroxysteroid dehydrogenase type 2 activity.¹⁶⁰ The latter response to this increment in dietary potassium due to added fruits and vegetables indicates greater glucocorticoid access to the mineralocorticoid receptor, which along with aldosterone, yields added stimulation of distal nephron potassium excretion.¹⁷² Importantly, potassium in plant-based food is predominantly in the form of non-chloride ions, a form that facilitates urine potassium excretion.¹⁷³ Additionally, CKD patients have increased potassium secretion into the gastrointestinal tract with increased fecal potassium losses.¹⁷⁴ Together, these data support that CKD patients have the capacity to increase potassium excretion in response to an increment in dietary potassium.

A comprehensive analysis of 36,000 CKD 3-4 patients reported a U-shaped mortality risk for both higher (> 5 mEq/L) and lower (< 3.5 mEq/L) serum potassium levels.¹⁷⁵ Additionally, a secondary analysis of 840 patients from the MDRD cohort found that higher urine potassium excretion, a surrogate marker for higher dietary potassium intake, was associated with a lower risk for all-cause mortality.¹⁷⁶ This higher potassium intake was unlikely due to pharmacologic potassium administration to these CKD patients known to have reduced ability to excrete potassium; more likely, the higher potassium intake was due to higher intake of potassium-containing foods like plant-based food components that these patients tolerated without untoward effects. In support of this supposition, patients with CKD stages 3-4 tolerated diets high in plant-based food components given to improve calcium/phosphate metabolism^{177,178} and to treat metabolic acidosis^{96,160} without developing hyperkalemia. In addition, patients in these studies tolerated these plant-based diets despite their taking inhibitors of the renin-angiotensin-aldosterone system (RAAS), which can limit urine potassium excretion, given for kidney protection. Limiting other drugs that can also reduce urine potassium excretion, such as nonsteroidal antiinflammatory drugs (NSAIDs) as was done in these

studies,^{96,160,177,178} might help CKD patients eating a potassium-rich diet avoid hyperkalemia.

The 2015 report of the American Dietary Guidelines Advisory Committee¹⁶⁹ recommends a daily potassium intake for the general population of adults of 4700 mg (121 mmol). Nevertheless, average daily potassium intake for US adults is 2155 mg (55 mmol), likely being lower than recommended due to poor intake of fruits and vegetables.¹⁶⁹ The NKF-KDOQI initiative has recommended potassium intake identical to the general population in patients with CKD stages 1 and 2 (> 4 g/day or > 102 mmol/day) and reduced intake of potassium to 2 to 4 g/day (51–102 mmol/day) in patients with CKD 3 and 4.¹⁷⁹ Because of the described benefits of plant-based diets, recommendations to restrict potassium should ideally be tailored to allow CKD patients 1) with hypertension to eat the DASH diet with its high plant-based content⁴² for hypertension control; 2) the added plant-based food components for effective management of the disturbed calcium/phosphate metabolism of CKD^{177,178}; and 3) with metabolic acidosis to receive base-producing fruits and vegetables for management of their metabolic acidosis.^{96,160} To date, published studies have not defined a clear role for potassium binding agents that might allow for liberalization of dietary potassium intake in patients with CKD, particularly when given with RAAS blockers.¹⁸⁰

- **Phosphorus.** Patients with CKD and reduced GFR are in positive phosphorus balance yet can maintain phosphorus levels in the normal range until advanced CKD stages (i.e., stages 4–5) through phosphaturia induced by increases in fibroblast growth factor-23 and parathyroid hormone.¹⁸¹ Phosphorus metabolism is linked to increase CVD, vascular calcifications, and morbidity in CKD patients.¹⁸² Nevertheless, dietary phosphate intake measured by 24-hour urine phosphate excretion was not associated with ESKD incidence or with cardiovascular mortality as long as the resulting serum phosphate levels were within the normal range.¹⁸³ Historically, practitioners have limited dietary phosphate intake by limiting intake of phosphate-containing foods, particularly processed foods that often have elemental phosphate-added preservatives.¹³³ The elemental phosphates added to processed foods are of particular concern because they are quickly and completely absorbed from the gastrointestinal tract when ingested.¹⁸⁴ By contrast, phosphate in plant-based food is in the form of phytates (phytic acid) that have much lower gastrointestinal tract absorption than the elemental phosphate in food additives.^{178,185} This helps explain the lower levels of serum phosphate and FGF23 in CKD patients eating a primarily plant-based diet in comparison to those eating a diet more typical of developed societies in which the protein source was primarily an animal-based based diet.^{177,178} In addition, CKD patients eating the plant-based diet had need for fewer phosphate-binding medications with a resulting lower pill burden in these patients with advanced CKD.¹⁷⁸

Together, these data support the recommendation for restricted phosphorus intake in CKD patients, specifically to less than 800 mg/day¹³³ in comparison to the average of 1000 mg/day for the average US adult.¹⁴⁰ Importantly, this requires limited or no intake of processed food that contains elemental phosphate additives. Relatedly, fresh

and home-prepared foods yielded better phosphate control than processed foods without compromising nutritional status.¹⁸⁶ Dietary phosphate restriction can also be helped by restricting intake of animal-based protein and/or by adding plant-based protein with its less absorbable phosphate^{178,185} and/or substituting plant-based for animal-based dietary protein.

- **Calcium.** Recommendations for daily dietary calcium intake are 1000 mg (23 mmol) for adults age 19 to 50 years, 1200 mg (28 mmol) for females 51 to 70 years old, and 1200 mg for all adults older than 70 years for the general population.¹⁸⁷ Published data are insufficiently robust to offer a specific recommendation for dietary calcium intake for CKD patients beyond that for the general population.¹⁸⁸ Recognizing the potential benefits of a plant-based compared to an animal protein-based diet in CKD patients, such diets affect overall calcium metabolism. Individuals eating a primary animal-based protein diet have higher urine calcium excretion than those eating a primary plant-based diet.¹⁸⁹ Earlier studies supported that this higher urine calcium excretion was due to bone buffering of acid produced from metabolism of the ingested animal-based protein with resorption of bone mineral and release of calcium into extracellular fluid.¹⁹⁰ Studies that are more recent support that a likely greater contributor to the increased urine calcium excretion observed with the diets high in animal-based protein is that such diets are typically very low in oxalate that binds to calcium in the gastrointestinal tract, thereby preventing calcium absorption and promoting its fecal excretion.¹⁹¹ Greater gastrointestinal calcium absorption promoted by these low oxalate diets leads to greater calcium transport into the extracellular fluid with greater increased urine calcium excretion.¹⁹² On the other hand, diets high in plant-based protein have high amounts of calcium-binding oxalate that yield decreased gastrointestinal calcium absorption with subsequent lower urine calcium excretion.^{191,192}
- **Dietary fiber.** Advanced CKD patients appear to have a “leaky gut” with translocation of endotoxins and bacterial DNA with untoward downstream untoward consequences including increased inflammation and increased CVD morbidity.¹⁹³ Diets typical of developed societies drive a milieu of metabolic abnormalities including uremic toxin production, inflammation, and immunosuppression that ultimately promotes progressive kidney failure and cardiovascular disease.^{193,194} Some dietary components are metabolized in the colon by fermentation, with the two predominant types being saccharolytic (carbohydrate) and proteolytic (protein). Saccharolytic fermentation is beneficial with generation of short chain fatty acids such as butyrate, propionate and acetate, whereas proteolytic fermentation produces uremic toxins such as p-cresyl sulfate and indoxyl sulphate.¹⁹⁵ Furthermore, diets with higher animal-based protein-to-fiber ratio are associated with higher serum levels of p-cresyl sulfate and indoxyl sulphate.¹⁹⁵ p-Cresol sulfate is a microbial metabolite that is found in urine and likely derives from secondary metabolism of p-cresol. Indoxyl sulfate, also known as 3-indoxylsulfate, is a metabolite of dietary l-tryptophan. Indole is produced from l-tryptophan in the human intestine via tryptophanase-expressing gastrointestinal bacteria. Whereas indoxyl is produced from indole via

enzyme-mediated hydroxylation in the liver. *SULT1A1* appears to be the primary sulfotransferase enzyme involved in the conversion of indoxyl into indoxyl sulfate in the liver. On the other hand, analysis of 14,543 NHANES participants demonstrated that increases in dietary fiber intake were associated with reductions in inflammation and mortality in patients with CKD.¹⁹⁶ Increased dietary fiber was associated with decreased mortality in CKD patients¹⁹⁶ and with reduced serum levels of creatinine and urea.¹⁹⁷ The many discussed benefits of increasing the dietary intake of plant-based dietary components in CKD patients include increasing their fiber intake that is generally lower than that of the general population.¹⁹⁷

Mean daily fiber intake in the general US population is 16 grams, much lower than the recommended daily intake of 25 to 30 grams.¹⁹⁸ On the other hand, mean daily fiber intake for CKD patients was 12 grams,¹⁹⁹ showing even lower intake than the general population, likely due to the historic limitation of plant-based foods in CKD patients. The American Heart Association recommends daily fiber intake greater than 25 grams to help reduce CVD risk.²⁰⁰ Because of the comparably high CVD risk in CKD patients,⁴ it seems prudent to follow this recommendation for patients with CKD. Increased intake of plant-based foods is an effective strategy to increase dietary fiber but this intervention should be limited to CKD patients who can tolerate the associated increased potassium load without developing hyperkalemia.

- **Micronutrients.** Unfortunately, there have been few published studies regarding the micronutrient status of CKD patients and their need for trace elements and vitamins required for energy production, cell and organ function, and for growth. Although some studies show that many hemodialysis patients have dietary intakes of trace elements and vitamins that fall below recommended values,²⁰¹ plasma levels of these substances in hemodialysis patients are typically not low.²⁰² Other studies in a general population show that high intake of several micronutrients was associated with a reduced CKD risk.²⁰³ Despite most guidelines that recommend multivitamin supplements for dialysis patients,¹³³ a comprehensive examination of the literature found insufficient evidence to support routine multivitamin use in hemodialysis patients.²⁰⁴ This is in stark contrast to the need for water-soluble vitamin supplementation when kidney replacement therapy is initiated, especially for folate.

Nevertheless, because both peritoneal and hemodialysis patients are routinely exposed to dialysates without trace elements and because many predialysis CKD patients are frequently prescribed diuretic agents, practitioners have understandable concerns that CKD patients are at risk for being deficient in water-soluble vitamins. Some studies report low vitamin C levels in peritoneal dialysis patients²⁰⁵ and vitamin C supplements reduced dosage requirements for erythropoiesis-stimulating agents in other studies.²⁰⁶ Still others report that L-carnitine supplement reduced resistance to erythropoietin-induced red blood cell production.²⁰⁷ Relatedly, half or more of nondialysis CKD patients had plasma studies consistent with iron deficiency.²⁰⁸ Accordingly, dietary guidelines for CKD patients often include supplement for the

water-soluble vitamins including B₁, B₆, B₁₂, C, folate, and niacin.¹³³ Vitamin D is commonly prescribed to CKD patients and its appropriate use is discussed elsewhere in this book. The paucity of published studies regarding the status of other fat-soluble vitamins in CKD patients do not allow an evidence-base recommendation for them.

Because of the disproportionately high contribution of coronary artery calcification (CAC) to the increased CVD risk of CKD²⁰⁹ and the contribution of vitamin K-dependent matrix Gla protein to the physiologic prevention of CAC,²¹⁰ vitamin K status of CKD patients has attracted attention. Dietary intake of vitamin K was low in hemodialysis patients in comparison to the general population^{211,212} and was inversely related to CVD-related mortality in non-dialysis CKD patients.²¹³ Because many fruits and vegetables contain vitamin K,²¹¹ their addition to the diets of CKD patients would appear to help provide vitamin K and possibly ameliorate CAC.

Regarding micronutrients in CKD, it seems that a reasonable approach would be to recommend the previously-described components of a “healthy” diet as tolerated, emphasizing plant-based proteins, with a smaller proportion of animal-based protein to provide micronutrients that are more abundant in animal-based compared to plant-based food components such as riboflavin.

OVERALL DIET CONSIDERATIONS IN THE MANAGEMENT OF CKD

DIETARY APPROACHES TO PREVENTION OF THE CONDITIONS THAT MOST COMMONLY CAUSE CKD IN DEVELOPED SOCIETIES, DIABETES AND HYPERTENSION

As indicated, prevention of CKD importantly focuses on prevention of the two major syndromes leading to CKD in developed societies, diabetes (predominantly T2D) and hypertension. Tables 60.1 and 60.2 list strategic considerations in this approach for T2D and hypertension, respectively, and tactics to execute these strategies. Strategies common to both T2D and hypertension include diets that reduce net endogenous acid production, contain a high proportion of plant-based components, limit dietary NaCl, and reduce the risk for obesity. Institution of a predominately plant-based diet can achieve each of these strategies common to prevention of T2D and hypertension. This common thread of a dietary approach designed to prevent both T2D and hypertension supports a population approach emphasizing increased intake of plant-based foods that holds promise to reduce the incidences of these two syndromes that contribute significantly to early death and increased disability in developed societies.¹ Such interventions, including this focus on increased intake of plant-based foods, are best applied at the level of overall populations if they are to reduce the incidences of T2D and hypertension and thereby CKD incidence. Doing so at the population level will likely require policy changes combined with public marketing. Practitioners can also consider these approaches for their individual patients deemed to be at increased risk for T2D and hypertension as suggested by family history, ethnic history, and/or their socioeconomic circumstance.

Table 60.3 Dietary Approaches to Reduce the Risk for Initiation of CKD in Susceptible Individuals

Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Emphasize an overall diet with a high ratio of fiber to animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize fats that enhance kidney health	Promote polyunsaturated fats, including omega 3's

Table 60.4 Dietary Approaches to Reduce the Risk for Progression of Established CKD

Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables as tolerated - Minimize animal-based foods - Minimize NaCl ("salt")
Emphasize an overall diet which yields a gut microbiome that promotes kidney health	- Promote plant-based foods as tolerated - Minimize animal-based foods, particularly red meat

DIETARY APPROACHES TO MINIMIZING THE RISK FOR INITIATION OF CKD IN SUSCEPTIBLE INDIVIDUALS/POPULATIONS

Practitioners can consider dietary approaches for individuals who presently have no CKD but are at increased risk for it (Table 60.3). Their increased risk might be because they have diabetes (T2D or T1D) and/or hypertension, have other systemic conditions associated with CKD, or because of a strong family history. Again, these strategies might be applied to overall populations considered to be at increased CKD risk, e.g., US ethnic minorities concentrated in low-income communities or Native Americans living on reservations.⁴ Primary care providers can also consider these interventions in patients that they manage with T2D and/or hypertension.

DIETARY APPROACHES TO MINIMIZING THE RISK FOR PROGRESSION OF ESTABLISHED CKD TO ESKD

Table 60.4 depicts approaches that practitioners might consider to reduce the risk for progression to ESKD for individuals with established CKD, particularly those with reduced eGFR at presentation. Successful prevention of, or delay of, ESKD adds to the quality of life to CKD patients and reduces their management costs.⁴ Practitioners currently underuse dietary strategies designed to prevent or slow CKD progression.⁹

Table 60.5 Dietary Approaches to Reduce Mortality in Established CKD

Strategy	Tactic
Emphasize an overall diet with a high ratio of fiber to animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Minimize dietary NaCl ("salt")	- Promote fresh fruits and vegetables as tolerated - Minimize processed foods

DIETARY APPROACHES TO REDUCING MORTALITY IN ESTABLISHED CKD

Practitioners might consider the approaches listed in Table 60.5 in an effort to reduce the mortality risk for individuals with established CKD, especially those with reduced eGFR because mortality in CKD is inversely associated with eGFR.¹²² Practitioners managing CKD patients often focus on CKD-related complications and less so on strategies to reduce mortality.¹²² Currently, when the management focus includes mortality reduction, the focus is typically on pharmacologic strategies⁴ with less focus on the effectiveness of dietary strategies.² As described for strategies to reduce CKD progression, dietary strategies designed to reduce mortality appear to be underused.

GENERAL DIETARY MANAGEMENT OF PATIENTS WITH CKD

Table 60.6 lists general management approaches to CKD patients, distinguished by important nutritional components.

DIETARY APPROACHES TO MANAGING SOME CKD-RELATED COMPLICATIONS

Table 60.7 lists some effective dietary strategies for management of the indicated CKD-related complications. Practitioners might use these dietary strategies along with appropriate pharmacologic interventions.

CONSIDERATIONS FOR ENABLING CKD PATIENTS TO EAT A MORE PLANT-BASED DIET

Many of the dietary recommendations for CKD patients supported by published studies include increasing the plant-based component of their diets. Nevertheless, diets typical of developed societies are comprised predominantly of animal-based components.³³ Facilitating CKD patients eating these diets requires practitioners, health system leaders, and even government leaders to address real and perceived barriers for patients to acquire and consume such diets, including expense (e.g., fresh fruits and vegetables are generally more expensive than packaged, processed foods) and accessibility (many low-income communities have limited access to fresh fruits and vegetables).²¹⁴ Once CKD patients gain access to these foods, practitioners must simplify the

Table 60.6 General Approach for a Dietary Recommendation for an Individual With CKD and Reduced GFR

Dietary Consideration	Recommendation
Energy	30-35 kcal/kg body weight/day 5%-15% from carbohydrates - Avoid fructose
Protein	0.7-0.8 g/ideal body weight/day for non-dialysis patients 1.2 g/ideal body weight/day for dialysis patients - Non-nitrogen keto-analogues might be substituted for a portion of recommended dietary protein - Plant-based protein can be considered for patients able to tolerate the additional potassium load
Sodium	<2g (90 mmol)/day unless otherwise contraindicated
Potassium	>4 g (102 mmol)/day for CKD stages 1 and 2 2-4 g (51-102 mmol)/day for CKD stages 3 and 4 - As instructed by provider for CKD stage 5
Phosphorous	< 800 mg/day - Promote plant-based sources over animal-based sources - Minimize processed foods
Calcium	1000 mg (23 mmol)/day for adults age 19-50 years 1200 mg (28 mmol)/day for adult females age 51-70 years 1200 mg (28 mmol)/day for adults >70 years
Fiber	≥25 g/day
Micronutrients	- Promote a primary plant-based diet - Multivitamin supplement for dialysis patients

Table 60.7 Dietary Interventions That Help in the Management of CKD-Related Complications

Strategy	Tactic
Nephropathy progression	- Reduce net endogenous acid production - Promote a “kidney friendly” gut microbiome
Hypertension	- Minimize NaCl (“salt”) - Promote plant-based food as tolerated - Potassium as tolerated - Magnesium as tolerated
Disturbed calcium/phosphate metabolism	- Promote plant-based food as tolerated - Minimize processed foods
Metabolic acidosis	- Promote base-producing fruits and vegetables as tolerated - Minimize animal-based protein - Minimize NaCl (“salt”)
Increased protein catabolism	Treat metabolic acidosis with the described dietary approaches

approach to healthy food patterns such that patients have a variety of food options to achieve food patterns that achieve intended goals.²¹⁵ This simplified approach must include culturally sensitive instructions for food preparation.⁹⁶ In addition, gaining support of all members of a patient's household helps facilitate successful dietary interventions for CKD patients whose recommended diet often contrasts with that of remaining household members.⁹⁶

FUTURE RESEARCH DIRECTIONS

Despite published data that support effective dietary approaches to the management of patients with CKD, many areas require further examination. For example, the benefits to CKD patients of a plant-based diet on mortality, progression of underlying nephropathy, and on some CKD-related comorbid conditions support such diets as first-line treatment of CKD patients with pharmacologic therapy as adjunctive.⁹ Future studies will determine, however, if this suggested approach will become standard of care for CKD management. In addition, future research will better determine in what circumstances these diets are most effective and how best to facilitate their implementation by CKD patients. In the meantime, it is imperative that clinicians, importantly those providing primary care, recognize the kidney protection benefits of the discussed dietary interventions for CKD patients.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

- Which of the following diet characteristics are associated with lower CKD-related mortality?
 - Low in red meat
 - High ratio of fiber to animal-based protein
 - High in fresh fruit and vegetables
 - Low in NaCl (salt)
 - All of the above

Answer: e

Rationale: A healthy diet in CKD reduces mortality and enhances length and quality of life of individuals with CKD.^{2,3}

- Describe dietary strategies to reduce accumulation of nitrogenous wastes that contribute to the complications of uremia.
 - Reduce dietary protein intake to minimum levels needed to maintain health
 - Substitute non-nitrogen ketoanalogs for nitrogen-containing amino acids
 - Treat underlying metabolic acidosis to reduce protein catabolism
 - Implement strategies to preserve glomerular filtration rate to enhance excretion of nitrogenous wastes
 - All of the above

Answer: e

Rationale: Symptoms of uremia are associated with breakdown products of protein catabolism. Therefore strategies to reduce accumulation of nitrogenous waste products include lower protein diets (0.6–0.8 g/bw/day), sometimes supplemented with non-nitrogen ketoanalogs.^{139,143,144} However, the largest study of the impact of overall dietary protein restriction (MDRD) was not conclusive.¹⁴⁶ A growing list of studies support the importance of the nature of protein (plant based Vs animal based) rather than the quantity of protein. Specifically, plant-based vs. animal-based dietary protein yields better outcomes for the CKD related complications of increased protein, catabolism, disturbed calcium/phosphate metabolism, and metabolic acidosis.

- Describe dietary strategies to consider when treating the disturbed calcium/phosphate metabolism of CKD.
 - High content of animal-based protein
 - High content of plant-based protein
 - High content of processed rather than fresh foods
 - Low intake of dietary phosphates in the form of phytates
 - High intake of elemental phosphate

Answer: b

Rationale: Patients with CKD have positive phosphorus balance and increased phosphorus metabolism is linked with increased CVD morbidity, vascular calcifications and CKD progression. Phosphate in plant based diets is in the form of “phytates” with lower gastrointestinal phosphorus absorption than additive phosphorus in processed food.^{178,182}

Increased animal protein diet is linked with increased urinary calcium excretion whereas plant based diets are associated with increased oxalate which lowers calcium absorption from gut and is linked with reduced urinary excretion.^{191,192}

- A 64-year-old African-American man with hypertension presented to the CKD clinic for evaluation. Hypertension was diagnosed 15 years ago and he is maintained on lisinopril 40 mg and amlodipine 5 mg daily. He is a non-smoker but does provide history of dialysis dependency in his mother.

- Blood pressure 132/80, weight 210 lbs (BMI 29); rest of his physical exam was unremarkable.
- Laboratory data:
 - Na⁺ 137 mEq/L
 - K⁺ 4.5 mEq/L
 - Cl⁻ 105 mEq/L
 - HCO₃⁻ 22 mEq/L
 - Cr 1.8 mg/dL (estimated GFR by the CKD EPI formula = 45 mL/min/1.73 m²)
 - Urine analysis shows 2+ protein by dipstick, with negative microscopic exam
 - Urine protein-to-creatinine ratio = 1000 mg/g creatinine
 - Arterial blood pH = 7.36, PCO₂ = 38 mm Hg

This patient has borderline low serum HCO₃⁻ level but has yet to develop frank metabolic acidosis to the degree that warrants alkali treatment by KDIGO guidelines (KDIGO recommends Na⁺-based alkali treatment for CKD patients with serum HCO₃⁻ < 22 mEq/L).

What are some food vs. medication factors to consider when contemplating initial management of, and even prevention of, metabolic acidosis in CKD?

- The low ratio of plant-based to animal-based protein in diets typical of developed societies contributes to metabolic acidosis in CKD.
- Addition/substitution of small amounts of plant-based protein can improve metabolic acidosis in CKD.
- Treating metabolic acidosis in CKD with plant-based protein can concomitantly address other CKD complications.
- Some CKD patients with very low eGFR might not be able to tolerate the added K⁺ of plant-based food.
- All of the above

Answer: e

Rationale: Metabolic acidosis in CKD is linked with increased protein catabolism, increased CKD progression, muscle mass reduction, and mineral-bone disease acceleration. Base-producing diets have shown to be effective in amelioration of metabolic acidosis but due consideration must be given to potassium content of foods given to patients with advanced CKD particularly in diabetic patients.^{156–161}

5. A 54-year-old Caucasian woman with history of T2D presented to the CKD clinic for evaluation. Her T2D was diagnosed 15 years ago and she is maintained on lisinopril 40 mg and glipizide 5 mg daily. She is a cigarette smoker but does not provide history of dialysis dependency in her family.

- BP 122/78, weight 180 lbs (BMI 32); rest of her physical exam was unremarkable.
- Laboratory Data:
 - Na^+ 139 mEq/L
 - K^+ 4.9 mEq/L
 - Cl^- 105 mEq/L
 - HCO_3^- 19 mEq/L
 - Cr 1.5 mg/dL (estimated GFR by the CKD EPI formula = 39 mL/min/1.73 m²)
 - UA 1+ protein
 - Urine protein-to-creatinine ratio = 500 mg/g creatinine
 - Arterial blood pH 7.32, PCO_2 38 mm/Hg

What are some options to consider with the goal in mind to reduce the risk that her already reduced estimated GFR will decrease further toward, or to, end-stage kidney disease?

- a. Limit dietary intake of animal protein, particularly red-meat
- b. Increase intake of foods that promote a microbiome that minimizes production of kidney toxic substances
- c. Increase intake of plant-based foods, including fruits and vegetables
- d. Treat her metabolic acidosis, possibly adding Na^+ -based alkali, limiting dietary NaCl (salt), and adding base-producing fruits and vegetables as tolerated
- e. All of the above

Answer: e

Rationale: The key emphasis to slow the progression of advanced CKD is to change the diet approach from animal-based regimens to “plant based” regimens. Plant-based diets not only promote a healthy gut microbiome, but has impact on decreasing net endogenous acid load, promotes weight loss, CVD disease reduction, and improvement in hypertension and diabetes management.^{4,9,64,96,160,162}